

Effectiveness of chlorine on the microbial reduction in agricultural water

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1. Preliminary setup

Run this script from within the R project - the code now uses relative paths which may not work otherwise.

Load Required Packages

```
# Load packages
library(here)
library(writexl)
library(tidyverse)
library(ggpubr)
library(nlme)
library(ggResidpanel)
library(performance)
library(emmeans)
```

```
library(marginaleffects)
library(multcomp)
library(multcompView)
```

```
sessioninfo::session_info(pkgs = c("attached"))
```

– Session info

```
setting  value
version  R version 4.4.2 (2024-10-31)
os       macOS Sequoia 15.1.1
system   x86_64, darwin20
ui       X11
language (EN)
collate  en_US.UTF-8
ctype    en_US.UTF-8
tz       America/New_York
date     2025-01-20
pandoc   3.2 @
/Applications/RStudio.app/Contents/Resources/app/quarto/bin/tools/x86_64/
(via rmarkdown)
```

– Packages

package	* version	date (UTC)	lib	source
dplyr	* 1.1.4	2023-11-17	[1]	CRAN (R 4.4.0)
emmeans	* 1.10.6	2024-12-12	[1]	CRAN (R 4.4.1)
forcats	* 1.0.0	2023-01-29	[1]	CRAN (R 4.4.0)
ggplot2	* 3.5.1	2024-04-23	[1]	CRAN (R 4.4.0)
ggpubr	* 0.6.0	2023-02-10	[1]	CRAN (R 4.4.0)
ggResidpanel	* 0.3.0	2019-05-31	[1]	CRAN (R 4.4.0)
here	* 1.0.1	2020-12-13	[1]	CRAN (R 4.4.0)
lubridate	* 1.9.4	2024-12-08	[1]	CRAN (R 4.4.1)
marginaleffects	* 0.24.0	2024-11-25	[1]	CRAN (R 4.4.1)
MASS	* 7.3-64	2025-01-04	[1]	CRAN (R 4.4.1)
multcomp	* 1.4-26	2024-07-18	[1]	CRAN (R 4.4.0)
multcompView	* 0.1-10	2024-03-08	[1]	CRAN (R 4.4.0)
mvtnorm	* 1.3-3	2025-01-10	[1]	CRAN (R 4.4.1)
nlme	* 3.1-166	2024-08-14	[1]	CRAN (R 4.4.2)
performance	* 0.13.0	2025-01-15	[1]	CRAN (R 4.4.1)
purrr	* 1.0.2	2023-08-10	[1]	CRAN (R 4.4.0)
readr	* 2.1.5	2024-01-10	[1]	CRAN (R 4.4.0)
stringr	* 1.5.1	2023-11-14	[1]	CRAN (R 4.4.0)
survival	* 3.8-3	2024-12-17	[1]	CRAN (R 4.4.1)
TH.data	* 1.1-2	2023-04-17	[1]	CRAN (R 4.4.0)
tibble	* 3.2.1	2023-03-20	[1]	CRAN (R 4.4.0)
tidyr	* 1.3.1	2024-01-24	[1]	CRAN (R 4.4.0)
tidyverse	* 2.0.0	2023-02-22	[1]	CRAN (R 4.4.0)
writexl	* 1.5.1	2024-10-04	[1]	CRAN (R 4.4.1)

```
[1] /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library
```

Set global options

```
options(contrasts = c('contr.sum', 'contr.poly'),
        dplyr.width = Inf, pillar.print_max = 50, pillar.print_min = 50)

knitr::opts_chunk$set(eval = T, echo = T, cache = T, fig.align = 'center')

set.seed(1484) #accroding to https://www.random.org
```

Define custom functions

The candidate models are Weibull and log-logistic. The derivation of the reparameterized log-logistic is provided at the end of this document.

```
# Weibull and reparameterized weibull
wb = function(x, log10n0, alpha, beta) {
  log10n0 - (x/exp(alpha))^beta
}

wbrepar = function(x, log10n0, TD, beta, D = 1){
  log10n0 - D*(x/TD)^beta
}

# Log-logistic and reparameterized log-logistic
lg = function(x, log10n0, kappa, sigma){
  log10n0 - log10(1+ exp((log(x)-kappa)/sigma^2))
}

lgrepar = function(x, log10n0, TD, sigma, D = 1){
  log10n0 - log10(1+ exp((log(x)- (log(TD) - log(10^D -
1)*sigma^2))/sigma^2))
}
```

2. Chlorine demand (CLD)

2.1 Data entry

```
chl = read.delim(here('Data (study 1)', "obj1_chl_NM.txt")) |>
  mutate(across(1:2, factor),
         sample = fct_relevel(sample, 'di', 'ag1', 'ag2', 'ag3', 'ag4'))
str(chl)

'data.frame':   49 obs. of  7 variables:
 $ sanitizer      : Factor w/ 2 levels "accutab","sanidate": 1 1 1 1 1 1 1 1
1 1 ...
```

```

$ sample          : Factor w/ 5 levels "di","ag1","ag2",...: 2 2 2 2 2 2 2 3
3 3 ...
$ chl.added       : int   2 4 10 15 20 30 40 2 4 10 ...
$ chl.measured.ave: num   0.85 2.87 8.23 11.77 16.4 ...
$ chl.measured.se : num   0.11 0.19 0.36 0.58 0.41 0.35 0.65 0.05 0.07 0.4
...
$ CLD.ave        : num   1.15 1.13 1.77 3.23 3.6 4.35 4.15 1.73 3.68 6.42
...
$ CLD.se         : num   0.11 0.19 0.36 0.58 0.41 0.35 0.65 0.05 0.07 0.4
...

```

```
saveRDS(chl, here('Results (study 1)', 'CLD_all_ave.RDS'))
```

```
# Grouping based on sanitizer for each Ag water
```

```
CLD_m1 = readRDS(here('Results (study 1)', "CLD_all_ave.RDS")) |>
  dplyr::filter(sample=="ag1" | sample=="ag2" |
                sample=="di" & sanitizer=="accutab")
```

2.2 Plot CLD

```
CLD_m1 = readRDS(here('Results (study 1)', "CLD_all_ave.RDS")) |>
  dplyr::filter(sample=="ag1" | sample=="ag2" |
                sample=="di" & sanitizer=="accutab") |>
  mutate(chl.added = factor(chl.added))
```

```
str(CLD_m1)
```

```

'data.frame':   21 obs. of  7 variables:
 $ sanitizer      : Factor w/ 2 levels "accutab","sanidate": 1 1 1 1 1 1 1 1
1 1 ...
 $ sample        : Factor w/ 5 levels "di","ag1","ag2",...: 2 2 2 2 2 2 2 3
3 3 ...
 $ chl.added     : Factor w/ 7 levels "2","4","10","15",...: 1 2 3 4 5 6 7 1
2 3 ...
 $ chl.measured.ave: num   0.85 2.87 8.23 11.77 16.4 ...
 $ chl.measured.se : num   0.11 0.19 0.36 0.58 0.41 0.35 0.65 0.05 0.07 0.4
...
 $ CLD.ave       : num   1.15 1.13 1.77 3.23 3.6 4.35 4.15 1.73 3.68 6.42
...
 $ CLD.se        : num   0.11 0.19 0.36 0.58 0.41 0.35 0.65 0.05 0.07 0.4
...

```

```
var_names = c(
  "sample" = "Samples",
  "ag1" = "Ag1",
  "ag2" = "Ag2",
  "di" = "DI water"
)
```

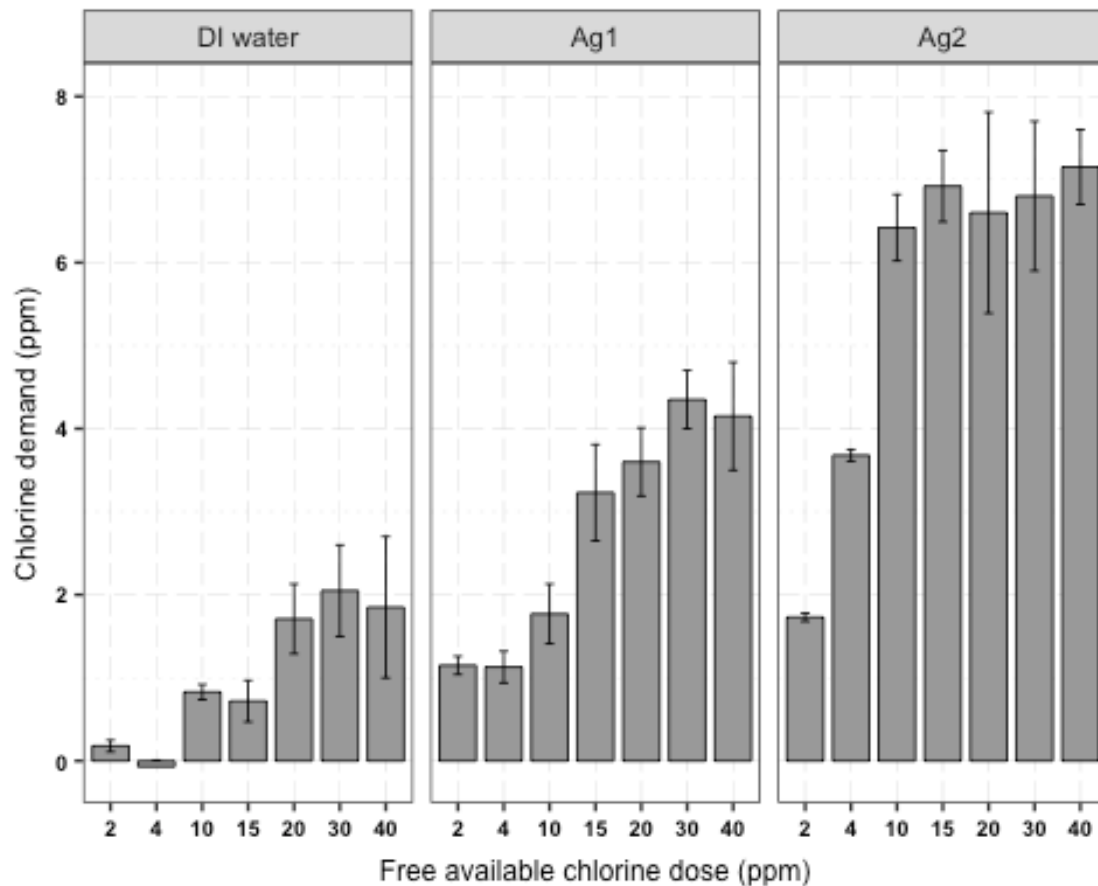
```
p_chl_demand = CLD_m1 |>
  filter (sample == "di" | sample=="ag1" | sample=="ag2") |>
```

```

ggplot (aes(x=chl.added, y=CLD.ave, fill=sample)) +
geom_bar (position = position_dodge(0.8),
          width=0.8,
          color ="black",
          stat="identity",
          linewidth=0.3) +
facet_grid (~ sample, labeller = as_labeller(var_names)) +
geom_errorbar(aes(ymin=CLD.ave-CLD.se,
                  max=CLD.ave+CLD.se),
              position = position_dodge(0.8),
              linewidth=0.3,
              width=0.2) +
labs (
  x = "Free available chlorine dose (ppm)",
  y = "Chlorine demand (ppm)",
  size = 9, face="bold") +
scale_fill_manual(values=c("grey60", "grey60", "grey60"),
                  name="",
                  breaks=c("di", "ag1", "ag2"),
                  labels=c("DI water", "Ag1", "Ag2")) +
ylim (-0.1, 8) +
theme_bw (base_family="") +
theme(
  text = element_text(family = "Arial"),
  legend.position = "none",
  plot.title = element_text(color="black", size=9, face = "bold", hjust =
0.5),
  axis.text.x=element_text(size = 7, color="black", face="bold", hjust =
0.5, vjust = 0.5),
  axis.text.y=element_text (size = 7, color="black", face="bold"),
  axis.title.x = element_text (size = 9, vjust = -1),
  axis.title.y = element_text (size = 9, vjust = 2),
  panel.grid.major = element_line (color="gray", linewidth=0.1,
linetype="longdash"),
  panel.grid.minor = element_line (color="lightgray", linewidth=0.1,
linetype="dotted"),
  panel.background = element_rect (fill = "transparent"),
  legend.background = element_rect (fill = "transparent", color =
"transparent"),
  plot.background = element_rect (fill = "transparent", color =
"transparent")) +
guides(
  color = guide_legend(override.aes = list(size = 5))
)

p_chl_demand

```



```
ggsave(file=here('Results (study 1)', "Fig_1 CLD_plot.tiff"),
  plot = p_chl_demand,
  width=9, height=3.9, units = c("cm"), dpi=600)
```

3. Control group

3.1 Data entry and basic calculations

```
m1c = readRDS(here('Data (study 1)', 'obj1_m1.rda')) |>
  filter(sample == "di" & time == 5) |>
  mutate(across(c(1, 4, 7:8, 10), factor))

m1c_w = m1c |>
  tidyr::spread (rep, survival)

# calculate mean, sd, and se
m1c_w$mean = apply(
  m1c_w[,c(11:12)], 1, base::mean, na.rm=TRUE) # calculate mean
m1c_w$sd = apply(
  m1c_w[,c(11:12)], 1, stats::sd, na.rm=TRUE) # calculate Std Dev
m1c_w$se = m1c_w$sd/sqrt(3) # calculate Std Error
```

```

m1c_w |>
  dplyr::select (sanitizer, sample, chl.measured, temp, time, microb, assay,
mean, sd, se) |>
  saveRDS(here('Results (study 1)', "m1c_w.RDS"))

# Log10 reduction for E coli 0 ppm to 2 ppm
ec_chl0 <- m1c |>
  filter(microb == "ec" & chl.measured == 0 & time=="5") |>
  dplyr::select(survival)
mean(ec_chl0$survival)

[1] 6.67

sd(ec_chl0$survival)/sqrt(3)

[1] 0.0473

ec_chl2 <- m1c |>
  filter(microb == "ec" & chl.measured == 1.8 & time=="5") |>
  dplyr::select(survival)
mean(ec_chl2$survival)

[1] 1.93

sd(ec_chl2$survival)/sqrt(3)

[1] 0.219

mean(ec_chl0$survival-ec_chl2$survival)

[1] 4.74

sd(ec_chl0$survival-ec_chl2$survival)/sqrt(3)

[1] 0.265

# Log reduction for TV 0 ppm to 4 ppm
tv_chl0 <- m1c |>
  filter(microb == "tv" & chl.measured == 0 & time=="5") |>
  dplyr::select(survival)
mean(tv_chl0$survival)

[1] 5.21

sd(tv_chl0$survival)/sqrt(3)

[1] 0.0624

tv_chl2 <- m1c |>
  filter(microb == "tv" & chl.measured == 1.8 & time=="5") |>
  dplyr::select(survival)
mean(tv_chl2$survival)

```

```

[1] 2.54
sd(tv_chl2$survival)/sqrt(3)
[1] 0.277
mean(tv_chl0$survival-tv_chl2$survival)
[1] 2.67
sd(tv_chl0$survival-tv_chl2$survival)/sqrt(3)
[1] 0.316
tv_chl4.1 <- m1c |>
  filter(microb == "tv" & chl.measured == 4.1 & time=="5") |>
  dplyr::select(survival)
mean(tv_chl4.1$survival)
[1] 1.66
sd(tv_chl4.1$survival)/sqrt(3)
[1] 0.192
mean(tv_chl0$survival-tv_chl4.1$survival)
[1] 3.55
sd(tv_chl0$survival-tv_chl4.1$survival)/sqrt(3)
[1] 0.248
# Max inactivation
mean(ec_chl0$survival) - 1
[1] 5.67
mean(tv_chl0$survival) - 0.347
[1] 4.86

```

3.2 Plot control group

```

m1c_w = readRDS(here('Results (study 1)', "m1c_w.RDS")) |>
  filter(time=="5")
m1c_w$chl.measured.fac = as.factor (round(m1c_w$chl.measured, 0))

# LOD for each microbe
lod_tv = 0.347
lod_ec = 1.0

p_tv = m1c_w |>
  filter(microb=="tv") |>
  ggplot (aes(x=chl.measured.fac, y=mean)) +

```



```

geom_bar(stat = "identity", position = position_dodge(0.1),
         linewidth = 0.3, fill = "darkgray", color = 'black') +
geom_errorbar (aes(x=chl.measured.fac, ymin=mean-se, ymax=mean+se),
              width=0.3, alpha=0.5, color = "black",
              position = position_dodge(0.1)) +
geom_hline (yintercept=lod_tv, linetype="dashed", color = "#00529b",
linewidth = 0.3) +
scale_shape_manual(values=c (3, 6)) +
ggtitle (expression(paste("Tulane virus")))) +
labs (y= bquote (Log[10]~PFU~per~ml)) +
ylim (0, 7) +
theme_bw (base_family="") +
theme(
  plot.title = element_text(color="black", size=9, face = "bold", hjust =
0.5),
  axis.text.x=element_text (size = 9, color="black", face="bold"),
  axis.text.y=element_text (size = 9, color="black", face="bold"),
  axis.title.x = element_blank(),
  axis.title.y = element_text (size = 9, vjust = 2),
  panel.grid.major = element_line (color="gray", linewidth=0.1,
linetype="longdash"),
  panel.grid.minor = element_line (color="lightgray", linewidth=0.1,
linetype="dotted"),
  panel.background = element_rect (fill = "transparent"),
  legend.background = element_rect (fill = "transparent", color =
"transparent"),
  plot.background = element_rect (fill = "transparent", color =
"transparent"))

p_ec = m1c_w |>
filter(microb=="ec" & chl.measured!="14.3") |>
ggplot (aes(x=chl.measured.fac, y=mean)) +
geom_bar(stat = "identity", position = position_dodge(0.1),
         linewidth = 0.2, fill = "darkgray", color = 'black') +
geom_errorbar (aes(x=chl.measured.fac, ymin=mean-se, ymax=mean+se),
              width=0.3, alpha=0.5, color = "black",
              position = position_dodge(0.1)) +
geom_hline (yintercept=lod_ec, linetype="dashed", color = "#00529b",
linewidth = 0.3) +
scale_shape_manual(values=c (3, 6)) +
ggtitle (expression(italic("E. coli")))) +
labs (y= bquote (Log[10]~CFU~per~ml)) +
ylim (0, 7) +
theme_bw (base_family="") +
theme(
  plot.title = element_text(color="black", size=9, face = "bold", hjust =
0.5),
  axis.text.x=element_text (size = 9, color="black", face="bold"),
  axis.text.y=element_text (size = 9, color="black", face="bold"),
  axis.title.x = element_blank(),

```

```

    axis.title.y = element_text (size = 9, vjust = 2),
    panel.grid.major = element_line (color="gray", linewidth=0.1,
linetype="longdash"),
    panel.grid.minor = element_line (color="lightgray", linewidth=0.1,
linetype="dotted"),
    panel.background = element_rect (fill = "transparent"),
    legend.background = element_rect (fill = "transparent", color =
"transparent"),
    plot.background = element_rect (fill = "transparent", color =
"transparent"))

combined_plot = ggpubr::ggarrange(p_ec,
                                p_tv, ncol = 2,
                                labels = c("A", "B"),
                                font.label = list(size=9, face="bold",
family="Arial"
                                ))
combined_plot <- annotate_figure(combined_plot,
                                bottom = text_grob("Free available chlorine
(ppm)",
                                                    size = 9,
                                                    family = "Arial"))

ggsave(plot=combined_plot,
        file= here('Results (study 1)', "Fig_2_control_group_plot.tiff"),
        width=9, height=5, units = c("cm"), dpi=600)

```

4. Modeling microbial inactivation kinetics

4.1 Data entry and approach

```

m1_file = here('Data (study 1)', 'obj1_m1_NM.txt')
m1 = read.table(file = m1_file, sep = '\t', header = T) |>
  mutate(across(c(1, 4, 7:8, 10), factor),
         temp_fac = factor(temp),
         time_fac = factor(time, labels = c('5min', '10min')),
         microb = fct_recode(microb, 'ec' = 'ecoli', 'tv' = 'tv'))

saveRDS(m1, here('Data (study 1)', 'obj1_m1.rda'))

```

Since there were insufficient time points included in the study to fit a primary model, time was initially included as a (categorical) fixed effect for the model parameters in the Weibull and log-logistic models. In all cases, however, neither time nor any interactions with time were statistically significant ($p > 0.05$), and thus it was determined that - given the impossibility of fitting a primary model and the clear indication that the overwhelming majority of inactivation was occurring, for all treatments and microbes, within the first 5 minutes - to refit the model using only the data from the 5min exposure.

4.2 Model Development and Evaluation

```
m1 = readRDS(here('Data (study 1)', 'obj1_m1.rda')) |>
  filter(sample != 'di' & time == 5) |>
  droplevels()

m1_mod_wb = gnls(survival ~ wb(x = chl.measured,
                             log10n0 = log10n0,
                             alpha = alpha,
                             beta = beta),
                 data = m1,
                 params = log10n0+alpha+beta ~ sample*microb,
                 start = list(log10n0 = c(6, rep(0, 3)),
                              alpha = c(5, rep(0, 3)),
                              beta = c(1, rep(0, 3))))

m1_mod_lg = gnls(survival ~ lg(x = chl.measured,
                              log10n0 = log10n0,
                              kappa = kappa,
                              sigma = sigma),
                 data = m1,
                 params = log10n0+kappa+sigma ~ sample*microb,
                 start = list(log10n0 = c(5, rep(0, 3)),
                              kappa = c(.1, rep(0, 3)),
                              sigma = c(.2, rep(0, 3))))
```

Initial comparison via AICc indicates log-logistic is the best fitting:

```
compare_performance(m1_mod_wb, m1_mod_lg, metrics = c('AICc'))
```

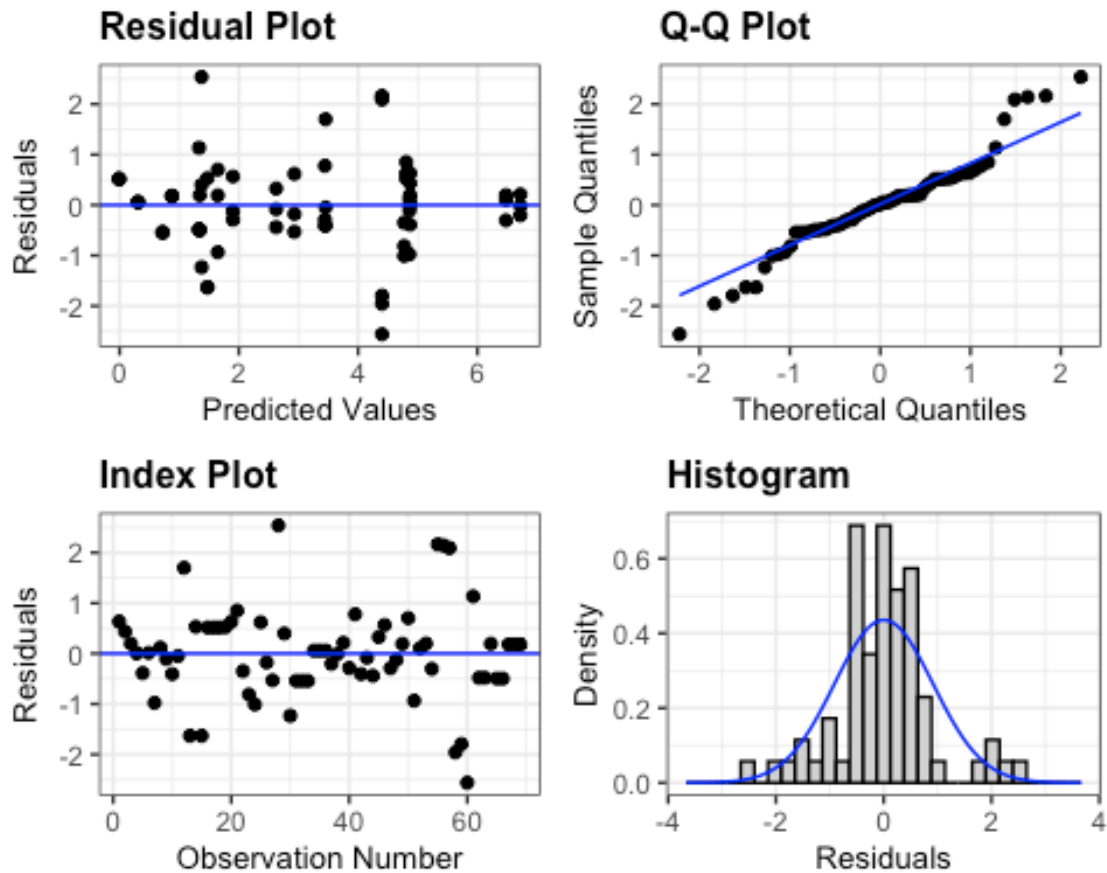
Comparison of Model Performance Indices

Name	Model	AIC (weights)	AICc (weights)	BIC (weights)	R2
m1_mod_wb	gnls	158.8 (0.325)	165.5 (0.325)	187.9 (0.325)	0.904
m1_mod_lg	gnls	157.4 (0.675)	164.0 (0.675)	186.4 (0.675)	0.906

Name	RMSE	Sigma
m1_mod_wb	0.634	0.697
m1_mod_lg	0.627	0.690

Examination of the residuals suggests some evidence of heteroscedasticity (although likely actually the result of censoring of observations below LOD). Nevertheless, the model is compared with heteroscedasticity for any improvement.

```
m1_res_lg = resid(m1_mod_lg, type = 'normalized')
m1_fit_lg = fitted(m1_mod_lg)
resid_auxpanel(m1_res_lg, m1_fit_lg)
```



Power model is best fitting.

```
m1_mod_lg_exp = update(m1_mod_lg, weights = varExp())
m1_mod_lg_pow = update(m1_mod_lg, weights = varPower(),
                        control = gnlsControl(maxIter = 100))
m1_mod_lg_ident = update(m1_mod_lg, weights = varIdent(form = ~1|microb))

compare_performance(m1_mod_lg, m1_mod_lg_exp, m1_mod_lg_pow,
                    m1_mod_lg_ident, metrics = c('AICc')) |>
  dplyr::select(Name, Model, AICc, AICc_wt)
```

Comparison of Model Performance Indices

Name	Model	AICc (weights)
m1_mod_lg	gnls	164.0 (0.130)
m1_mod_lg_exp	gnls	165.9 (0.049)
m1_mod_lg_pow	gnls	160.4 (0.768)
m1_mod_lg_ident	gnls	165.8 (0.053)

Refit the model using the re-parameterized version of the model. The code below converts the estimates in the original model into the corresponding estimates in the reparameterized model, so that they can be used as starting values.

```
# Extract design matrix for each treatment combination and name the rows
# accordingly
E = model.matrix(~ sample*microb, data = m1) |>
  unique()
ids = as.numeric(rownames(E))
rownames(E) = t(apply(m1[ids, c('sample', 'microb')], 1, paste0, collapse = '
'))

# Get ordering for marginal means and make sure the rows in the "E" matrix
match
#emmeans(m1_mod_lg, ~ sample:microb, param = 'kappa')
E = E[c(3, 4, 1, 2),]

# calculate marginal means for each treatment combination
kappa_emms = emmeans(m1_mod_lg_pow, ~ sample:microb, param = 'kappa')|>
  as.data.frame() |>
  dplyr::select(emmean) |>
  unlist()

n0_emms = emmeans(m1_mod_lg_pow, ~sample:microb, param = 'log10n0')|>
  as.data.frame() |>
  dplyr::select(emmean) |>
  unlist()

sigma_emms = emmeans(m1_mod_lg_pow, ~sample:microb, param = 'sigma') |>
  as.data.frame() |>
  dplyr::select(emmean) |>
  unlist()

# Calculate the time to 3 log10 reductions implied by the fitted estimates
# for each treatment combination
td_emms = log(10^3-1)*sigma_emms + kappa_emms

# Then, calculate the coefficients (based on the E matrix) implied by those
# 3 log10 reductions
td_coefs = round(solve(E)%%td_emms, 1)
n0_coefs = round(solve(E)%%n0_emms, 1)
sigma_coefs = round(solve(E)%%sigma_emms, 1)

# These values should be extremely close to the fitted values in the
# reparameterized model

m1_mod = gnls(survival ~ lgrepar(x = chl.measured,
                                log10n0 = log10n0,
                                TD = TD,
```

```

        sigma = sigma,
        D = 3),
    data = m1,
    params = log10n0+TD+sigma ~ sample*microb,
    weights = varPower(),
    start = list(log10n0 = n0_coefs,
                 TD = td_coefs,
                 sigma = sigma_coefs))

saveRDS(m1_mod, here('Results (study 1)', "m1_mod.RDS"))

# Extract estimates and their 95% CI
m1_mod_coef = coef(m1_mod)
m1_mod_confint = confint(m1_mod)

as.data.frame (cbind(
  Coefs = names(m1_mod_coef),
  Est = round (as.numeric(m1_mod_coef), 3),
  lwr = round (m1_mod_confint[,1], 3),
  upr = round (m1_mod_confint[,2], 3)
)) |>
write_xlsx(here('Results (study 1)', "m1_mod_summary.xlsx"))

readxl::read_xlsx(here('Results (study 1)', "m1_mod_summary.xlsx"))

# A tibble: 12 × 4
  Coefs          Est    lwr    upr
  <chr>         <chr> <chr> <chr>
1 log10n0.(Intercept)  5.719  5.25  6.189
2 log10n0.sample1      0.085 -0.385 0.555
3 log10n0.microb1      0.879  0.409  1.349
4 log10n0.sample1:microb1 0.036 -0.434 0.506
5 TD.(Intercept)      8.547  6.455 10.638
6 TD.sample1          3.36   1.269  5.452
7 TD.microb1          -7.844 -9.936 -5.753
8 TD.sample1:microb1  -3.462 -5.553 -1.371
9 sigma.(Intercept)    0.603  0.538  0.668
10 sigma.sample1      -0.038 -0.103  0.027
11 sigma.microb1       0.136  0.071  0.201
12 sigma.sample1:microb1 0.089  0.024  0.154

```

4.3 Estimation, Inference, and Plotting

Only microbe species has a significant effect on the initial population (log10n0) parameter, while there is a significant interaction effect between microbe and sample on the chlorine required for log 3 reduction (TD parameter), and for the sigma parameter (which characterizes the variation in chlorine susceptibility within the microbial population).

```
m1_mod = readRDS(here('Results (study 1)', "m1_mod.RDS"))
```

```
(m1_ftest_n0 = joint_tests(m1_mod, param = 'log10n0'))
```

model term	df1	df2	F.ratio	p.value
sample	1	55	0.130	0.7240
microb	1	55	13.440	0.0010
sample:microb	1	55	0.020	0.8810

```
(m1_ftest_TD = joint_tests(m1_mod, param = 'TD'))
```

model term	df1	df2	F.ratio	p.value
sample	1	55	9.900	0.0026
microb	1	55	54.000	<.0001
sample:microb	1	55	10.500	0.0020

```
(m1_ftest_sigma = joint_tests(m1_mod, param = 'sigma'))
```

model term	df1	df2	F.ratio	p.value
sample	1	55	1.300	0.2583
microb	1	55	16.780	0.0001
sample:microb	1	55	7.200	0.0096

```
write_xlsx(list(log10n0 = as.data.frame(m1_ftest_n0),
              TD = as.data.frame(m1_ftest_TD),
              sigma = as.data.frame(m1_ftest_sigma)),
           path = here('Results (study 1)',
                       'm1_param_ftests.xlsx'))
```

Estimates and contrasts for the different parameters:***

```
(m1_emm_n0 = emmeans(m1_mod, pairwise ~ microb, param = 'log10n0'))
```

```
$emmeans
```

	microb	emmean	SE	df	lower.CL	upper.CL
ec		6.60	0.413	55	5.77	7.43
tv		4.84	0.243	55	4.35	5.33

Results are averaged over the levels of: sample
Confidence level used: 0.95

```
$contrasts
```

	contrast	estimate	SE	df	t.ratio	p.value
ec	- tv	1.76	0.479	55	3.670	0.0006

Results are averaged over the levels of: sample

```
(m1_emm_TD = emmeans(m1_mod, pairwise ~ microb:sample, param = 'TD'))
```

```
$emmeans
```

	microb	sample	emmean	SE	df	lower.CL	upper.CL
ec	ag1		0.60	0.65	55	-0.70	1.90

tv	ag1	23.21	2.43	55	18.33	28.09
ec	ag2	0.80	0.56	55	-0.31	1.92
tv	ag2	9.57	3.40	55	2.75	16.38

Confidence level used: 0.95

```
$contrasts
contrast      estimate    SE df t.ratio p.value
ec ag1 - tv ag1  -22.61 2.52 55  -8.980 <.0001
ec ag1 - ec ag2   -0.20 0.85 55  -0.240 0.9950
ec ag1 - tv ag2   -8.97 3.46 55  -2.590 0.0580
tv ag1 - ec ag2   22.41 2.50 55   8.970 <.0001
tv ag1 - tv ag2   13.64 4.18 55   3.260 0.0100
ec ag2 - tv ag2   -8.76 3.45 55  -2.540 0.0640
```

P value adjustment: tukey method for comparing a family of 4 estimates

```
(m1_emm_sigma = emmeans(m1_mod, pairwise ~ microb:sample, param = 'sigma'))
```

```
$emmeans
microb sample emmean    SE df lower.CL upper.CL
ec      ag1    0.790 0.1140 55    0.561    1.019
tv      ag1    0.340 0.0252 55    0.289    0.390
ec      ag2    0.688 0.0413 55    0.605    0.770
tv      ag2    0.594 0.0473 55    0.499    0.689
```

Confidence level used: 0.95

```
$contrasts
contrast      estimate    SE df t.ratio p.value
ec ag1 - tv ag1    0.450 0.1170 55    3.850 0.0020
ec ag1 - ec ag2    0.102 0.1220 55    0.840 0.8340
ec ag1 - tv ag2    0.196 0.1240 55    1.590 0.3950
tv ag1 - ec ag2   -0.348 0.0484 55   -7.190 <.0001
tv ag1 - tv ag2   -0.254 0.0536 55   -4.740 <.0001
ec ag2 - tv ag2    0.094 0.0628 55    1.490 0.4490
```

P value adjustment: tukey method for comparing a family of 4 estimates

4.4 Plot 3-log10 reduction (TD)

```
m1_emm2_TD = emmeans(m1_mod, pairwise ~ sample:microb,
                      param = 'TD')
```

```
sample_labs = c(ag1 = 'Ag1', ag2 = 'Ag2')
```

```
m1_ch3d_plot = m1_emm2_TD |>
  cld(Letters = letters, reversed = T, adjust = 'none') |>
  as.data.frame() |>
  mutate(.group = gsub(' ', '', .group),
```

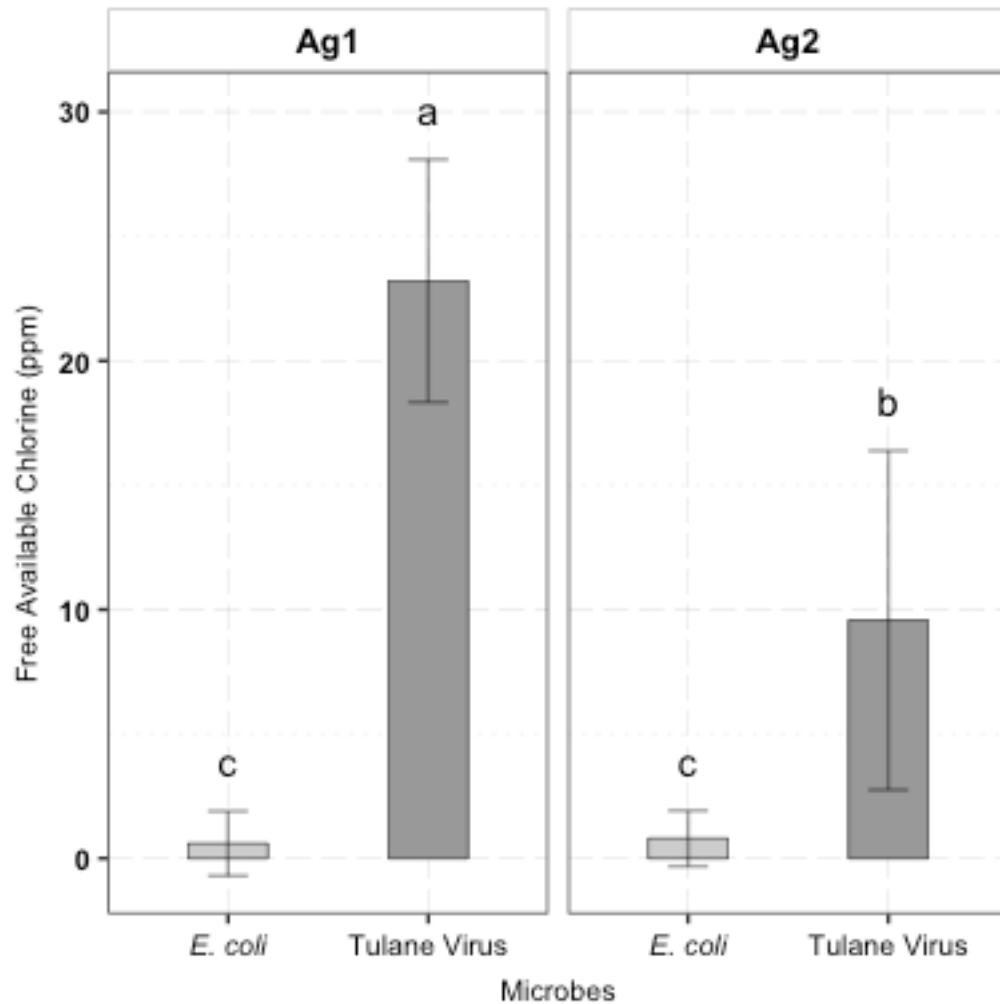


```

    Microbe = factor(microb, labels = c('E. coli', 'Tulane Virus')) ) |>
  ggplot(aes(x = Microbe, fill = Microbe))+
  facet_grid(cols = vars(sample), labeller = as_labeller(sample_labs))+
  geom_col(aes(y = emmean), show.legend = F, color = "black", linewidth =
0.2, width = 0.4) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.2,
alpha=0.4)+
  geom_text(aes(y = upper.CL+2, label = .group)) +
  scale_y_continuous('Free Available Chlorine (ppm)')+
  scale_x_discrete('Microbes', labels = c(expression(italic("E. coli")),
"Tulane Virus"))+
  scale_fill_manual(values = c("E. coli" = "gray80", "Tulane Virus" =
"gray60")) +
  theme(
    plot.title = element_text(color="black", size=8, face = "bold", hjust =
0.5),
    axis.text.x=element_text(size = 8, color="black", face="bold", hjust = 0.5,
vjust = 0.5),
    axis.text.y=element_text (size = 8, color="black", face="bold"),
    axis.title.x = element_text (size = 8, vjust = -1),
    axis.title.y = element_text (size = 8, vjust = 2),
    panel.grid.major = element_line (color="gray", linewidth=0.1)
  )

```

m1_ch3d_plot



```
ggsave(plot=m1_ch3d_plot, file=here('Results (study 1)',
  "m1_ch3d_plot.tiff"),
  width=14, height=7, units = c("cm"), dpi=600)
```

4.5 Plot microbial inactivation

```
m1_grid = expand.grid(sample = levels(m1$sample),
  microb = levels(m1$microb),
  chl.measured = seq(0, 36, 0.5)) |>
  filter(!(microb == 'ec' & chl.measured > 13.5))

m1_preds = avg_predictions(m1_mod, newdata = m1_grid,
  by = c('sample', 'microb', 'chl.measured'))

microb_labs = c(ec = expression(italic('E. coli')), tv = 'Tulane Virus')

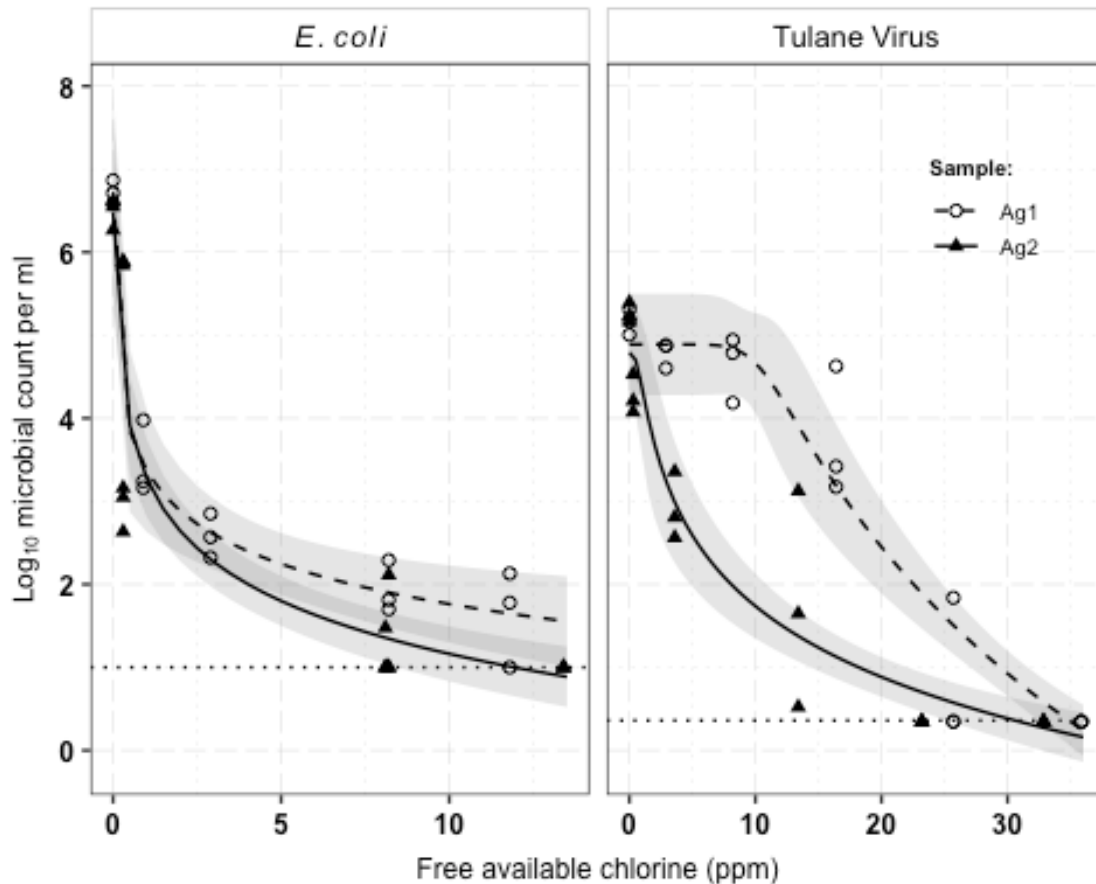
p_m1= m1_preds |>
  mutate(microb = factor(microb, labels = c('italic(E.~coli)',
```

```

'Tulane~Virus')),
  LOD = ifelse(microb == 'Tulane~Virus', 0.36, 1)) |>
  ggplot(aes(x = chl.measured)) +
  facet_grid(cols = vars(microb), labeller = label_parsed, scales = 'free_x')
+
  geom_hline(aes(yintercept = LOD), color = 'black', linetype = 3)+
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, group = sample), alpha =
.2,
            fill = 'grey45') +
  geom_line(aes(y = estimate, linetype = sample), color = "black") +
  geom_point(aes(y = survival, shape = sample),
            data = mutate(m1, microb = factor(microb,
            labels = c('italic(E.~coli)',
            'Tulane~Virus'))),
            size = 1.5) +
  guides (fill = guide_legend(title = 'Sample:'),
          color = guide_legend(title = 'Sample:'),
          linetype = guide_legend(title = 'Sample:'),
          shape = guide_legend(title = 'Sample:')) +
  labs(x = "Free available chlorine (ppm)",
       y = bquote(Log[10]~microbial~count~per~ml)) +
  scale_fill_manual(labels = c('Ag1', 'Ag2')) +
  scale_shape_manual(values = c(1, 17),
                    labels = c('Ag1', 'Ag2')) +
  scale_linetype_manual(values = c("dashed", "solid"),
                      labels = c('Ag1', 'Ag2')) +
  theme(text = element_text(family = "Arial"),
        legend.position = 'inside', legend.position.inside = c(.88, .8),
        legend.direction = "vertical",
        legend.text = element_text(size = 7),
        legend.title = element_text(size = 7, face = "bold"),
        legend.key.height = unit(0.4, "cm"),
        plot.title = element_text(color = "black", size = 9, face = "bold",
hjust = 0.5),
        axis.text.x = element_text(size = 9, color = "black", face = "bold",
hjust = 0.5, vjust = 0.5),
        axis.text.y = element_text(size = 9, color = "black", face = "bold"),
        axis.title.x = element_text(size = 9, vjust = -1),
        axis.title.y = element_text(size = 9, vjust = 2),
        panel.grid.major = element_line(
          color = "gray", linewidth = 0.1, linetype = "longdash"),
        panel.grid.minor = element_line(
          color = "lightgray", linewidth = 0.1, linetype = "dotted"),
        panel.background = element_rect(fill = "transparent"),
        legend.background = element_rect(fill = "transparent", color =
"transparent"),
        plot.background = element_rect(fill = "transparent", color =
"transparent"))

```

p_m1



```
ggsave(file = here('Results (study 1)', "Fig_3_trend_plot.tiff"),
        plot = p_m1,
        width=14, height=7, units = c("cm"), dpi=600)
```

These are model-based estimates of mean (along with their standard error) of the log10 PFU for each treatment combination.

```
m1_tab_grid = m1 |>
  dplyr::select(sample, microb, chl.added) |>
  rename(chl.measured = chl.added) |>
  distinct()
m1_ests = avg_predictions(m1_mod, newdata = m1_tab_grid,
                          by = c('microb', 'sample', 'chl.measured'))
m1_ests_tab = m1_ests |>
  as.data.frame() |>
  rename(Sample = sample, Microbe = microb) |>
  mutate(across(4:5, as.numeric),
         estimate = paste0(sprintf("%.2f", estimate),
                             '\u00b1 ', sprintf("%.2f", std.error)),
         chl.measured = paste0(chl.measured, ' ppm')) |>
  dplyr::select(Microbe, Sample, chl.measured, estimate) |>
  pivot_wider(names_from = chl.measured, values_from = estimate,
```

```

names_prefix = '')
print(m1_ests_tab)

# A tibble: 4 × 10
  Microbe Sample `0 ppm`      `2 ppm`      `4 ppm`      `10 ppm`      `15 ppm`
  <fct>   <fct>   <chr>        <chr>        <chr>        <chr>        <chr>
1 ec      ag1     6.72 ± 0.59 2.88 ± 0.26 2.40 ± 0.19 1.76 ± 0.24 1.48 ± 0.29
2 ec      ag2     6.48 ± 0.58 2.64 ± 0.19 2.00 ± 0.16 1.16 ± 0.17 0.79 ± 0.19
3 tv      ag1     4.89 ± 0.31 <NA>        4.89 ± 0.31 4.66 ± 0.31 <NA>
4 tv      ag2     4.79 ± 0.37 <NA>        2.86 ± 0.35 1.74 ± 0.20 <NA>
  `20 ppm`      `30 ppm`      `40 ppm`
  <chr>        <chr>        <chr>
1 <NA>        <NA>        <NA>
2 0.53 ± 0.21 <NA>        <NA>
3 2.45 ± 0.28 0.93 ± 0.14 -0.16 ± 0.19
4 0.88 ± 0.12 0.38 ± 0.13 0.03 ± 0.17

write_xlsx(m1_ests_tab,
           path = here('Results (study 1)',
                       'Estimated Conc. for 3 Log Reduction.xlsx'))

```

5. Reparameterizing the Log-Logistic Model

$$\log_{10}(N_t) = \log_{10}(N_0) - \log_{10}\left(1 + e^{\frac{\log(x) - \kappa}{\sigma^2}}\right)$$

Letting C_3 be the concentration at which $\log_{10}(N_t)$ is 3 below $\log_{10}(N_0)$, it follows that:

$$\log_{10}(N_0) - 3 = \log_{10}(N_0) - \log_{10}\left(1 + e^{\frac{\log(C_3) - \kappa}{\sigma^2}}\right)$$

Subtracting $\log_{10}(N_0)$ and multiplying by -1 gives:

$$\log_{10}\left(1 + e^{\frac{\log(C_3) - \kappa}{\sigma^2}}\right) = 3$$

$$1 + e^{\frac{\log(C_3) - \kappa}{\sigma^2}} = 10^3$$

$$e^{\frac{\log(C_3) - \kappa}{\sigma^2}} = 10^3 - 1$$

$$\frac{\log(C_3) - \kappa}{\sigma^2} = \log(10^3 - 1)$$

$$\log(C_3) - \kappa = \log(10^3 - 1)\sigma^2$$

$$\kappa = \log(C_3) - \log(10^3 - 1)\sigma^2$$

And substituting this back into the original form of the model gives:

$$\log_{10}(N_t) = \log_{10}(N_0) - \log_{10} \left(1 + e^{\frac{\log(x) - (\log(C_3) - \log(10^3 - 1)\sigma^2)}{\sigma^2}} \right)$$